

SOCIAL EMBLUE AI

Product Requirements Document

v3.1 — Social Listening Edition

10 Tools · 14 AI Agents · 30 TypeScript Files · 2 Dashboards

May 2026 | Confidential

Version	Date	Key Change	Status
v3.1	May 2026	Tool 1 completely rewritten — proactive keyword-based Social Listening + Historical Search	CURRENT
v3.0	May 2026	TypeScript conversion, TikTok auto, Post URL campaigns, Figma tokens	Superseded
v2.0	April 2026	Tool 10 Engage the Engagers, 14 agents, two dashboards	Superseded
v1.0	March 2026	Original 9-tool spec	Superseded

1. What Changed in v3.1

One critical correction that changes the meaning of the entire platform

CRITICAL CORRECTION

Tool 1 (Social Listening) was incorrectly specified in all previous versions. It was described as classifying comments on the client's own posts. That is wrong. Social Listening means monitoring the entire public internet for conversations around keywords the client specifies — including competitor names, industry terms, pain phrases, and product categories. This version corrects that specification completely.

The corrected Tool 1 has two distinct operating modes:

- Mode A — Real-Time Keyword Monitoring: Ongoing surveillance. System checks every 15 minutes for new public mentions of the client's keywords across all platforms. New mentions surface live in the Search Inbox feed.
- Mode B — Historical Search: On-demand batch analysis. Client specifies keywords and a date range (e.g. January 1 to December 31 last year). System pulls all matching public conversations from that period, classifies them, builds a volume chart, and extracts structured insights.

This single correction changes the value proposition of the entire platform. Social Emblue AI is no longer just a tool for managing a brand's own social media replies. It is a competitive intelligence and market listening platform that surfaces what the market is saying — giving brands the ability to enter conversations they were never part of before.

2. Tool 1 — Advanced Social Listening (Full Spec)

Proactive keyword monitoring + Historical search across the public social web

Social Listening is the act of monitoring the entire public internet — not just the brand's own posts — for conversations that are relevant to the brand. The client defines what is relevant by submitting keywords. The platform finds those conversations and surfaces them for action.

2.1 Mode A — Real-Time Keyword Monitoring

A MODE

Real-Time Monitoring

System continuously scans all configured platforms for new mentions of the client's keywords. Runs every 15 minutes. New mentions appear in the Search Inbox feed as they are detected. Operator can immediately engage a conversation from the feed.

How it works step by step:

1. Client submits keyword list in Settings (e.g. "GTBank complaint", "cheap flights Lagos", "where to buy iPhone Lagos")
2. Client selects which platforms to monitor and sets alert thresholds (urgency level, intent types)
3. Every 15 minutes: Apify actors search each platform for new public posts/comments containing the keywords
4. Agent 1 (Claude Haiku) classifies every result: sentiment, intent, urgency score, topic tags
5. Results appear in Search Inbox ranked by urgency score — highest opportunity at the top
6. Operator clicks any conversation → Agent 4 generates a reply → brand joins the conversation
7. High-urgency matches (purchase_intent + score 4-5) trigger instant SSE alert to War Room

2.2 Mode B — Historical Search

B MODE

Historical Search

Client selects keywords and a specific date range. System performs a one-time batch pull of all matching public conversations from that period. Results include a volume chart (mentions over time), sentiment breakdown, top conversations, topic clusters, and a full insights report.

Use cases:

- *"Show me everything said about GTBank customer service in 2024 — I want to understand the full year's conversation before we pitch them"*
- *"What were people saying about cheap flights in the last 6 months — our airline client"*

wants to know the pain points"

- "Compare conversation volume around 'MTN data' in Q1 vs Q3 last year"

What the output contains:

- Volume chart — mentions per day/week/month across the date range
- Sentiment breakdown — positive/neutral/negative split across the period
- Platform breakdown — where the conversation is loudest (X vs Reddit vs YouTube etc.)
- Topic clusters — what subtopics dominate the conversation
- Top conversations — highest-engagement posts/comments from the period
- Insights report — Agent 10 summary: FAQs, pain points, opportunities

2.3 Platform Coverage and Historical Depth

Platform	Real-Time	Historical	Max Lookback	Data Source	Monthly Cost
X / Twitter	Every 15 min	Full archive	7+ years	X API Basic (\$200/mo) or Apify	\$200 or \$29-249 (Apify)
Reddit	Every 15 min	Full archive	Years	PRAW (free) + Apify	\$0 (PRAW)
YouTube	Every 15 min	Full archive	Years	YouTube Data API v3 (free quota)	\$0
Instagram	Every 15 min	Limited	Few months	Apify only — no public keyword API	Included in Apify plan
TikTok	Every 15 min	Limited	Few months	Apify only — no public keyword API	Included in Apify plan
Facebook	Limited	Very limited	Recent public pages	Apify — most restricted	Included in Apify plan

NOTE

X (Twitter) and Reddit offer the best historical data. If a client wants a full year of keyword history, X and Reddit will return the richest results. Instagram and TikTok historical reach is limited to a few months due to platform restrictions on public data access.

2.4 Keyword Configuration

Each brand can configure multiple keyword groups. A keyword group is a named set of related search terms that are monitored together.

Field	Description	Example
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Field	Description	Example
Group name	Label for this set of keywords	Competitor Monitoring
Keywords	Search terms — can be phrases or single words	["GTBank complaint","GTBank down","GTBank useless"]
Platforms	Which platforms to search for these keywords	Instagram, X, Reddit
Mode	Real-time, Historical, or Both	Both
Date range (historical)	Start and end date for historical search	Jan 1 2024 to Dec 31 2024
Alert threshold	Urgency score at which to send live alert	Score >= 4
Intent filter	Only alert on specific intents	purchase_intent, complaint
Is active	Whether real-time monitoring is currently running	true / false

3. Search Inbox Screen Specification

The UI that surfaces all keyword-matched conversations

The Search Inbox is the screen where all keyword-matched conversations appear — both real-time and historical results. It is separate from the AI Reply Engine which handles the brand's own post comments.

3.1 Screen Layout

Zone	Content
Top bar	Page title "Social Listening" + "New Search" button + active keyword groups count
Filter strip	Filter by: keyword group, platform, date range, sentiment, intent, urgency
Left panel	Keyword group manager — list of all configured groups, add/edit/delete, active/paused toggle
Centre panel	Results feed — conversation cards ranked by urgency score, newest or highest score first
Right panel	Conversation detail — full post/comment text, author info, platform, engagement metrics, Agent 4 reply options
Bottom bar	Pagination or infinite scroll, total results count, export to CSV button

3.2 New Search Form (Historical Mode)

Field	Specification
Keywords input	Multi-tag input — user types a keyword and presses Enter. Each keyword becomes a removable chip. Min 1, max 20 keywords per search.
Platforms	Checkbox group: Instagram, X/Twitter, TikTok, Reddit, YouTube, Facebook. At least 1 required.
Mode selector	Radio: Real-time monitoring (ongoing) OR Historical search (date range)
Date range	Shown only when Historical is selected. Date picker: From and To. Max range: 24 months. Cannot be future dates.
Run button	Primary blue. Label: "Start Monitoring" (real-time) or "Run Historical Search" (historical). Disabled if no keywords or platforms selected.
Estimated time	Shown below button for historical: "Estimated completion: 2-5 minutes for 12 months of data"

3.3 Conversation Card (Results Feed)

Element	Specification
Platform badge	Coloured pill: IG pink, X black, TikTok dark, Reddit orange, YouTube red
Handle + timestamp	@author_handle · platform · time ago or exact date for historical
Matched keyword	Small chip showing which keyword triggered this result — e.g. "GTBank complaint"
Message preview	First 120 characters of the post/comment text. Matched keyword highlighted in blue.
Sentiment chip	Positive (green) / Neutral (grey) / Negative (red)
Intent chip	Inquiry / Complaint / Praise / Purchase Intent / Objection
Urgency indicator	Coloured dot: 5=red, 4=orange, 3=amber, 2=grey, 1=grey-light
Engagement metrics	Likes, replies, shares, views (where available from platform)
Action buttons	"View & Reply" — opens right panel with Agent 4 reply generation

3.4 Historical Search Results View

When a historical search completes, the Search Inbox switches to a results view with additional analytics panels:

Panel	Content
Summary bar	Total mentions found · Date range · Platforms searched · Search completed timestamp
Volume chart	Bar or line chart showing mention count per day/week/month across the date range. Colour coded: positive mentions in green, negative in red, neutral in grey. Hovering shows exact count for that period.
Sentiment donut	Circular chart: positive/neutral/negative split with percentages
Platform breakdown	Horizontal bar chart: which platform had the most mentions
Peak moment card	Identifies the single highest-volume day/week in the range with a reason if identifiable
Topic clusters	Top 5-10 topic clusters from Agent 2 — what subtopics dominate the conversation
Insights report	Agent 10 output: top FAQs, pain points with severity, opportunities identified
Results feed	All matching conversations below the analytics, same card format as real-time

Panel	Content
Export button	Download all results as CSV — handle, platform, date, text, sentiment, intent, urgency

4. New Database Tables for Social Listening

Three new tables added to `schema_full.sql`

4.1 keyword_groups

One row per named keyword group configured by a brand.

Column	Type	Description
group_id	BIGSERIAL PRIMARY KEY	Auto-generated ID
brand_id	INTEGER — FK to brands	Which brand owns this group
name	TEXT NOT NULL	Display name e.g. "Competitor Monitoring"
keywords	TEXT[] NOT NULL	Array of keyword strings
platforms	TEXT[] NOT NULL	Array of platform names to search
mode	TEXT — realtime historical both	Which mode this group uses
date_from	DATE	Start date for historical search (null if real-time only)
date_to	DATE	End date for historical search (null if real-time only)
alert_urgency_threshold	INTEGER DEFAULT 4	Minimum urgency score to trigger live alert
alert_intents	TEXT[]	Intent types that trigger alerts e.g. ["purchase_intent", "complaint"]
is_active	BOOLEAN DEFAULT TRUE	Whether real-time monitoring is running
created_at	TIMESTAMPTZ DEFAULT now()	When this group was created
last_run_at	TIMESTAMPTZ	When the most recent search ran

4.2 search_runs

One row per historical search execution. Preserved permanently so clients can revisit past searches.

Column	Type	Description
run_id	BIGSERIAL PRIMARY KEY	Auto-generated ID
brand_id	INTEGER — FK to brands	Which brand ran this search

Column	Type	Description
group_id	BIGINT — FK to keyword_groups	Which keyword group was used (null if ad-hoc)
keywords	TEXT[]	Exact keywords searched (snapshot at run time)
platforms	TEXT[]	Exact platforms searched
date_from	DATE	Start of search range
date_to	DATE	End of search range
mode	TEXT — realtime historical	Search mode
status	TEXT — pending running complete failed	Current run status
total_results	INTEGER DEFAULT 0	Total matching conversations found
positive_count	INTEGER DEFAULT 0	Count of positive sentiment results
negative_count	INTEGER DEFAULT 0	Count of negative sentiment results
neutral_count	INTEGER DEFAULT 0	Count of neutral sentiment results
peak_date	DATE	Date with highest mention volume in range
peak_count	INTEGER	Mention count on peak date
insights_summary	TEXT	Agent 10 plain-English summary
created_at	TIMESTAMPTZ DEFAULT now()	When the run was initiated
completed_at	TIMESTAMPTZ	When the run finished
error_msg	TEXT	Error message if status = failed

4.3 search_results

Every individual conversation found by any search run. Classification results stored alongside raw data.

Column	Type	Description
result_id	BIGSERIAL PRIMARY KEY	Auto-generated ID
run_id	BIGINT — FK to search_runs	Which run found this result
brand_id	INTEGER — FK to brands	Which brand owns this

Column	Type	Description
group_id	BIGINT — FK to keyword_groups	Which keyword group matched
matched_keyword	TEXT	The specific keyword that triggered this result
platform	TEXT — platform_enum	Which platform this came from
text	TEXT NOT NULL	Full text of the post or comment
author_handle	TEXT	Username of the author
author_id_ext	TEXT	Platform-native author ID (for potential outreach)
url	TEXT	Direct link to the original post/comment
posted_at	TIMESTAMPTZ	When the original post was published
sentiment	TEXT — positive neutral negative	Agent 1 classification
intent	TEXT — intent_enum	Agent 1 classification
urgency_score	INTEGER 1-5	Agent 1 urgency rating
topics	TEXT[]	Agent 1 topic tags (max 3)
likes	INTEGER DEFAULT 0	Engagement metric from platform
replies_count	INTEGER DEFAULT 0	Engagement metric from platform
shares	INTEGER DEFAULT 0	Engagement metric from platform
views	INTEGER DEFAULT 0	Engagement metric from platform
engaged	BOOLEAN DEFAULT FALSE	Whether the brand has replied to this conversation
created_at	TIMESTAMPTZ DEFAULT now()	When this result was stored

4.4 search_volume

Pre-computed volume data for the historical search volume chart. One row per time period per run.

Column	Type	Description
volume_id	BIGSERIAL PRIMARY KEY	Auto-generated ID
run_id	BIGINT — FK to search_runs	Which run this volume belongs to

Column	Type	Description
period_start	DATE	Start of this period (day/week/month)
period_end	DATE	End of this period
period_type	TEXT — day week month	Granularity of the period
mention_count	INTEGER DEFAULT 0	Total mentions in this period
positive_count	INTEGER DEFAULT 0	Positive mentions in this period
negative_count	INTEGER DEFAULT 0	Negative mentions in this period
neutral_count	INTEGER DEFAULT 0	Neutral mentions in this period

5. Updated Agent & Tool Mapping

How agents now connect to the corrected Tool 1

Agent	Tool	Previous Role (v3.0)	Corrected Role (v3.1)
Agent 1 — Agent_Listening	Tool 1	Classify comments on brand's own posts	Classify all keyword-matched public conversations from both real-time monitoring and historical search
Agent 2 — Agent_Clustering	Tool 2	Cluster brand's own comments	Cluster keyword search results into topic groups — what subtopics dominate the conversation around each keyword
Agent 3 — Agent_ContentStrategist	Tool 2	Strategy from clusters	Content strategy based on what the market is saying about the keyword — not just what the brand's followers say
Agent 10 — Agent_CommentMiner	Tool 8	Mine brand's own comments	Mine search results for FAQs, pain points, and opportunities — generates the Insights Report for historical searches
Agent 4 — Agent_Reply	Tool 3	Reply to brand's own comments	Also generates replies when operator chooses to engage a keyword-matched conversation from the Search Inbox
Agent 11 — Agent_CommandCentre	Tool 9	Monitor brand's own campaign health	Also monitors keyword volume spikes — a sudden surge in negative mentions around a keyword triggers a War Room alert

KEY INSIGHT

The brand's own post comments (Tool 3 AI Reply Engine) and the keyword-matched public conversations (Tool 1 Social Listening) are two completely separate data streams. They use the same agents but different data sources and different UI screens. Social Listening is proactive outreach. The Reply Engine is reactive management.

6. What Social Listening Enables — Business Value

Why this is the most strategically powerful feature in the platform

6.1 Competitive Intelligence

A bank client can monitor what people are saying about competitor banks — complaints, praises, switching signals. When someone tweets "GTBank just froze my account again, thinking of switching to Access Bank" — that is a live purchase-intent signal for Access Bank. Social Emblue AI surfaces it within 15 minutes.

6.2 Trend Detection Before It Peaks

Historical search reveals when a topic first emerged and how fast it grew. A client preparing a product launch can see whether the problem their product solves is growing in conversation volume. Agent 2 clustering identifies which sub-topics are accelerating.

6.3 Crisis Early Warning

A brand monitoring their own name as a keyword will detect a conversation spike before it becomes a viral complaint. A sudden 3x increase in negative mentions on a Tuesday afternoon — War Room turns AMBER before the brand's PR team notices.

6.4 Content Strategy From Real Audience Language

Agent 3 reads the keyword search results and extracts the exact phrases the audience uses. A content team is no longer guessing what language resonates — they are using words extracted from real conversations happening in the wild.

6.5 New Business Pitching

An agency using Social Emblue AI can run a historical search on a prospect brand's name before the sales meeting. Arriving at a pitch with "In 2024 there were 4,847 conversations about your brand. 68% were negative. The top complaint was response time. Here is what your customers are saying word for word" — that is a fundamentally different pitch than a generic social media proposal.

7. Apify Actor Mapping for Keyword Search

Which Apify actor handles each platform

Platform	Apify Actor	Supports Historical	Date Filter Support
Instagram	apify/instagram-hashtag-scraper or apify/instagram-search-scraper	Limited (few months)	Partial — recent posts only
X / Twitter	quacker/twitter-scraper or apify/twitter-scraper	Yes — up to years	Full from/to date filter
TikTok	clockworks/tiktok-scraper	Limited (few months)	Partial — recent videos
Reddit	trudax/reddit-scraper	Yes — years via Pushshift	Full date range filter
YouTube	streamers/youtube-scraper	Yes — years	Full date range filter
Facebook	apify/facebook-posts-scraper	Limited (public pages only)	Very limited

MONTH 2 ACTIVATION

Apify requires the `APIFY_API_TOKEN` environment variable and a paid plan (\$29-249/mo). The Starter plan (\$29/mo) gives enough credits for 1-3 brands. For a historical search covering 12 months across 4 platforms for one keyword, expect to use approximately \$5-10 in Apify credits.

7.1 How Apify Search Works in the Code

The existing `apifySync.ts` file handles CSV scraping. A new function is needed specifically for keyword search:

Function	Purpose
<code>runKeywordSearch(brandId, keywords, platforms, dateFrom, dateTo)</code>	Entry point — dispatches to platform-specific scrapers
<code>searchInstagramKeyword(keyword, dateFrom, dateTo)</code>	Runs <code>apify/instagram-hashtag-scraper</code> with keyword and date params
<code>searchXKeyword(keyword, dateFrom, dateTo)</code>	Runs <code>quacker/twitter-scraper</code> with keyword and since/until params
<code>searchRedditKeyword(keyword, dateFrom, dateTo)</code>	Runs <code>trudax/reddit-scraper</code> with keyword and date range
<code>searchYouTubeKeyword(keyword, dateFrom, dateTo)</code>	Runs <code>streamers/youtube-scraper</code> with keyword and date params

Function	Purpose
dateFrom, dateTo)	
searchTikTokKeyword(keyword, dateFrom, dateTo)	Runs clockworks/tiktok-scraper with hashtag/keyword param
buildVolumeChart(results, granularity)	Groups results by day/week/month to build volume chart data
saveSearchResults(runId, results)	Persists all results to search_results + search_volume tables

8. New API Routes for Social Listening

Routes to add to routes/api.ts or a new routes/listening.ts

Method + Route	Description	Auth
POST /api/v1/listening/keyword-groups	Create or update a keyword group for a brand	JWT required
GET /api/v1/listening/keyword-groups/:brand_id	List all keyword groups for a brand	JWT required
DELETE /api/v1/listening/keyword-groups/:group_id	Delete a keyword group	JWT required
POST /api/v1/listening/keyword-groups/:group_id/toggle	Pause or resume real-time monitoring	JWT required
POST /api/v1/listening/search	Run a new historical or real-time search — returns run_id immediately, processes async	JWT required
GET /api/v1/listening/runs/:brand_id	List all search runs for a brand	JWT required
GET /api/v1/listening/runs/:run_id/results	Get paginated results for a specific run	JWT required
GET /api/v1/listening/runs/:run_id/volume	Get volume chart data for a historical run	JWT required
GET /api/v1/listening/runs/:run_id/status	Check if a search run is still processing	JWT required
GET /api/v1/listening/feed/:brand_id	Get real-time keyword monitoring feed (all active groups combined)	JWT required
POST /api/v1/listening/results/:result_id/engage	Mark a search result as engaged and generate a reply via Agent 4	JWT required

9. Full Platform Tool Summary (v3.1)

All 10 tools with corrected descriptions

Tool	Name	What It Actually Does	Agent
01	Advanced Social Listening	Monitors the entire public internet for keyword-matched conversations. TWO modes: (A) Real-time — scans every 15 min for new mentions. (B) Historical — pulls all conversations for a keyword from a specified date range with volume chart, sentiment breakdown, and topic clusters.	Agent 1 (Haiku)
02	Search & Clustering	Groups keyword search results into meaningful topic clusters with opportunity scores. Identifies which subtopics dominate the conversation around each keyword. Generates specific content recommendations based on what the market is actually saying.	Agent 2 + 3 (Sonnet)
03	AI Reply Engine	Manages replies to comments on the brand's own posts. Generates 3 reply options per message with different tones. Confidence scoring determines auto-fire vs manual approval queue. Full audit trail of every reply sent.	Agent 4 (Sonnet)
04	Comment → DM Funnel	Converts public comments on the brand's posts into private DM conversations. Multi-step sequences with rate limiting and throttling. Works on Instagram and Facebook.	Agent 5 (Sonnet)
05	Social Response Dashboard	Aggregates all platform activity into a unified KPI dashboard. Listening score, reply score, funnel score, risk events. Weekly PDF reports to clients.	Agent 6 (Sonnet)
06	Attribution & Link Tracking	Creates unique tracked short links per reply, template, and campaign. Full attribution: Comment → Reply → Click → Booking. Revenue attributed to specific conversations.	Agent 8 (Logic)
07	Creative Performance Predictor	Scores any caption A-F before posting. Breakdown: comment/save/share/reach potential. Ranked next actions with before/after examples. Complete rewrite + 3 alt hooks.	Agent 9 (Sonnet)
08	Comment Mining & Insights	Processes all comments (own posts + keyword results) into structured intelligence. Ranked FAQ list, pain points with severity, objections with counter-responses, plain-English summary.	Agent 10 (Haiku + Sonnet)
09	Campaign Command Centre	Live War Room for campaign health. Four-level indicator: GREEN/AMBER/RED/CRISIS. Also monitors keyword volume spikes. Priority queue, crisis alerts, SSE live feed.	Agent 11 (Sonnet)
10	Engage the	Sends personalised replies to everyone who engaged	Agents

Tool	Name	What It Actually Does	Agent
	Engagers	a post. Three modes: live comment engagement, post URL batch campaign, manual trigger. TikTok now fully automated. Platform allocation sliders.	12+14+4

10. Build Status — v3.1

What is complete, what is new work required from the v3.1 correction

Component	Status	Notes	Priority
apifySync.ts — keyword search functions	NOT BUILT	New functions needed: runKeywordSearch, searchXKeyword, searchRedditKeyword, searchYouTubeKeyword, searchInstagramKeyword, searchTikTokKeyword	P0 — core feature
routes/listening.ts	NOT BUILT	9 new API routes for keyword groups, search runs, results, volume chart	P0 — core feature
keyword_groups table	NOT BUILT	Add to schema_full.sql	P0 — required
search_runs table	NOT BUILT	Add to schema_full.sql	P0 — required
search_results table	NOT BUILT	Add to schema_full.sql	P0 — required
search_volume table	NOT BUILT	Add to schema_full.sql	P0 — required
buildVolumeChart() function	NOT BUILT	Groups results by day/week/month for the chart	P0 — required for historical
Search Inbox screen (v-search-inbox)	HTML stub only	Full screen with keyword group panel, results feed, historical analytics, volume chart needs building	P0 — operator-facing
New Search form	NOT BUILT	Keyword input, platform selector, mode toggle, date range picker	P0 — required
Volume chart component	NOT BUILT	Bar/line chart showing mention volume over time from search_volume table	P0 — required
Keyword group manager UI	NOT BUILT	Left panel in Search Inbox — add/edit/delete groups, active toggle	P1
All 10 existing backend agents	COMPLETE	No changes required — Agent 1 classification logic is platform-agnostic	Done
All existing routes (api, auth, campaigns, realtime, dashboard)	COMPLETE	No changes required	Done
All 30 TypeScript source files	COMPLETE	0 type errors	Done
Engage the Engagers (Tool	COMPLETE	TikTok auto, post URL, allocation all	Done

Component	Status	Notes	Priority
10)		built	